

Taylor Nye

Washington University Medical School, St. Louis • (231) 871-1182 • nyet@wustl.edu

EDUCATION

University of Michigan Ph.D., Molecular, Cellular and Developmental Biology (MCDB) Dissertation: <i>Mechanisms of DNA Modification-Dependent Regulation in Gram-Positive Bacteria</i> (GPA 4.0/4.0) Advisor: Lyle Simmons, Ph.D.	Ann Arbor, MI August 2020
University of Michigan M.S., Bioinformatics (GPA 4.0/4.0)	Ann Arbor, MI August 2020
University of Michigan B.S., Microbiology Highest Honors and Distinction Undergraduate Thesis: <i>Identification of 5-methylcytidine and N6-methyladenosine DNA modifications in the Bacillus subtilis genome</i> (GPA 3.73/4.0)	Ann Arbor, MI May 2015

RESEARCH EXPERIENCE

Washington University in St. Louis <i>Senior Research Scientist</i> <i>Molecular Microbiology and Microbial Pathogenesis (MMMP)</i> Create and implement a novel molecular biology approach to identify proteins bound to locus of interest in bacteria	St. Louis, MO September 2025 – Present
Washington University in St. Louis <i>Postdoctoral fellow, MMMP</i> - Supervise and manage international collaboration with biopharmaceutical start-up Quretech Biology in Umea, Sweden to create new class of synthetic antibiotics active against multidrug-resistant bacteria - Lead team of physicians, clinical coordinators and lab technicians to develop a third-generation sequencing pipeline to detect antibiotic resistance genes in uropathogens from long-term catheterized patients	St. Louis, MO September 2020 – September 2025
University of Michigan <i>NSF Predoctoral Fellow, MCDB</i> - Developed expertise in bacterial genomic and transcriptomic techniques, bioinformatics, prokaryotic cell biology, and biochemistry - Identified DNA modifications and characterized physiological consequences in bacteria	Ann Arbor, MI July 2015 - August 2020

University of Michigan

Undergraduate Researcher, MCDB

Ann Arbor, MI

February 2013 – July 2015

- Identified the DNA methylome of the bacterium *B. subtilis*

TEACHING EXPERIENCE

Washington University in St. Louis

Postdoctoral lecturer

St. Louis, MO

Summer 2020

- Designed and taught WashU Specific Aims Immersion Class: Mechanisms of host-pathogen interactions as a blueprint for novel therapeutics

University of Michigan

Graduate Student Instructor, MCDB

Ann Arbor, MI

Sep 2019 – Dec 2019

- Taught 3 sections of Introductory Molecular Biology (90 min each; every week; ~25 students per section) covering various topics in Biology (macromolecules, enzyme kinetics, respiration/photosynthesis etc.)

PUBLICATIONS

- 19) Bazán-Villicaña J, Puri D, **Nye TM**, Hurto RL, Hazen J, Gualberto NC, Lopatto EDB, Pinkner JS, Dodson KW, Freddolino L, Hultgren SJ. The lac operon facilitates host glycan utilization and robust intracellular bacterial community development by uropathogenic *Escherichia coli*. (2026) **Proc. Natl. Acad. Sci. U.S.A.** In Review.
- 18) Tükenmez H*, **Nye TM***, Singh P*, Mychack A, Bonde M, Hickerson S, Obernuefemann CLP, Pinkner, JS, Zhu S, Sarkar S, Bharate JB, van der Lingen I, Ntuyenm AQ, Bhaskara Rao VU, Lindgren AEG, Klein H, Zou Z, Dodson KW, Walker S, Mateus A, Johansson J, Caparon MG, Almquist F, Hultgren SJ. Tunable TriPcides suppress virulence factor secretion during MRSA infection and kill dormant cells. (2026) **Sci Advances**. In Revision.
- 17) Hibma J, Munson J, Joshua D, **Nye TM**, Koutmou K, Simmons LA. TylA is a 23S and 16S 2'-O-methylcytidine methyltransferase important for ribosome assembly in *Bacillus subtilis*. (2026) **Nucleic Acids Res.** doi: 10.1093/nar/gkaf1531
- 16) Timm MR, Tamadonfar KO, **Nye TM**, Pinkner JS, Dodson KW, Ellebedy AH, Hultgren SJ. Vaccination with *Acinetobacter baumannii* adhesin Abp2D provides protection against catheter-associated urinary tract infection. (2025) **Nat Comm.** doi: 10.1038/s41467-025-62402-9
- 15) Zou Z, Singh P, Pinkner JS, Obernuefemann CLP, Xu W, **Nye TM**, Dodson KW, Almquist F, Hultgren SJ, Caparon MG. Dihydrothiazolo ring-fused 2-pyridone antimicrobial compounds treat *Streptococcus pyogenes* skin and soft tissue

infection. (2024) **Sci Adv.** doi: 10.1126/sciadv.adn7979

- 14) **Nye TM***, Zou Z*, Obernuefemann CLP, Pinkner JS, Lowry E, Kleinschmidt K, Bergeron K, Klim A, Flores-Mireles A, Walker JN, Wong DG, Desai A, Caparon MG, Hultgren SJ. (2024) Microbial co-occurrences on catheters from long-term catheterized patients. **Nat Comm.** doi: 10.1038/s41467-023-44095-0
- 13) Schroeder JW, Hurto RL, Randall JR, Wozniak KJ, Timko TA, **Nye TM**, Wang JD, Freddolino L, Simmons LA. (2023) RNaseH genes cause distinct impacts on RNA:DNA hybrid formation and mutagenesis genome-wide. **Sci Adv.** doi: 10.1126/sciadv.adi5945.
- 12) Fernandez NL, Chen Z, Fuller DEH, van Gitjenbeek LA, **Nye TM**, Biteen JS, Simmons LA. (2022) DNA methylation and RNA-DNA hybrids regulate the single-molecule dynamics of a DNA methyltransferase in *Bacillus subtilis*. **mBio.** doi: 10.1128/mbio.03185-22.
- 11) **Nye TM***, Tükenmez H*, Singh P*, Flores-Mireles AL, Obernuefemann CLP, Pinkner JS, Sarkar S, Bonde M, Lindgren AEG, Dodson KW, Johansson J, Almqvist F, Caparon MG, Hultgren SJ. (2022) New class of ring-fused 2-pyridone antibiotics disarm antibiotic resistance and have bactericidal activity against non-dividing enterococcal cells. **Proc. Natl. Acad. Sci. U.S.A.** doi:10.1073/pnas.2210921199
- 10) Tükenmez H*‡, **Nye TM*‡**, Bonde M, Caparon MG, Almqvist F, Hultgren SJ, Johansson J. (2022) Complete genome sequence of the uropathogenic methicillin-resistant *Staphylococcus aureus* strain, MRSA-1369. **Microbiol Resour Announc.** doi: 10.1128/mra.00981-22
- 9) McLean EK, **Nye TM**, Lowder FC, Simmons LA. (2022) The impact of RNA-DNA hybrids on genome integrity in bacteria. **Annu Rev Microbiol.** doi: 10.1146/annurev-micro-102521-014450.
- 8) Amemiya HA, Goss TJ, **Nye TM**, Hurto R, Simmons LA, Freddolino L. (2021) Distinct heterochromatin-like domains promote transcriptional memory and silence parasitic genetic elements in bacteria. **EMBO J.** 41(3):e108708. doi:10.15252/embj.2021108708.
- 7) **Nye TM**, McLean EK, Burrage AM, Kearns DB, Simmons LA. (2020) ZpdC is a plasmid-borne RNase HI that contributes to genome maintenance in the ancestral strain *Bacillus subtilis* NCIB 3610. **Mol. Microbol.** doi:10.1111/mmi.14601.
- 6) **Nye TM**, Fernandez N, Simmons LA. (2020) A positive perspective on DNA methylation: Regulatory functions of DNA methylation outside of host defense in Gram-positive bacteria. **Crit Rev Biochem Mol Biol.** 55(6): 576-591. doi:10.1080/10409238.2020.1828257.

- 5) **Nye TM**, van Gijtenbeek LA, Stevens AG, Schroeder JW, Randall JR, Matthews LA, Simmons LA. (2020) Methyltransferase DnmA is responsible for genome-wide N6-methyladenosine modifications at non-palindromic recognition sites in *Bacillus subtilis*. **Nucleic Acids Res.** Pii: gkaa266.
- 4) **Nye TM**, Jacob KM, Holley EK, Nevarez JM, Dawid SR, Simmons LA, Watson ME Jr. (2019) DNA methylation from a Type I restriction modification system influences gene expression and virulence in *Streptococcus pyogenes*. **PLoS Pathog.** 15(6): e1007841.
- 3) Randall JR, **Nye TM***, Wozniak KJ*, Simmons LA. (2019) RNase HIII is important for Okazaki fragment processing in *Bacillus subtilis*. **J Bacteriol.** Pii: JB.00686-18.
- 2) **Nye TM**, Schroeder JW, Kearns DB, Simmons LA. (2017) Complete Genome Sequence of Undomesticated *Bacillus subtilis* Strain NCIB 3610. **Genome Announc.** May 18;5(20). pii: e00364-17. doi: 10.1128/genomeA.00364-17
- 1) Burby PE*, **Nye TM***, Schroeder JW, Simmons LA. (2016) Implementation and Data Analysis of Tn-seq, Whole-Genome Resequencing, and Single-Molecule Real-Time Sequencing for Bacterial Genetics. **J Bacteriol.** 199(1). pii: e00560-16.

**these authors contributed equally, †corresponding author, indicates mentee*

INVITED LECTURES

GmPcides: Combatting the rise of antibiotic-resistant pathogens. Keynote speaker at QureTech Biology Annual Meeting, Umea, Sweden. 2022.

DNA methylation from a Type I restriction modification system influences gene expression and virulence in *Streptococcus pyogenes*, Eastern Michigan Seminar Series, Ypsilanti, Michigan. 2019.

Biology 140: Stimulating Science Seminar. Spring Arbor University. 2018.

PRESENTATIONS

Oral: Locus-specific identification of DNA-bound regulatory proteins in bacteria with CaTNAPP, Molecular Genetics of Bacteria and Phages, Madison, Wisconsin. 2025.

Poster: Combatting the rise of multidrug-resistant Gram-positive pathogens with novel ring-fused 2-pyridone antibiotics, International Conference on Enterococci, Porto, Portugal. 2022.

Poster: Combatting the rise of multidrug-resistant Gram-positive pathogens with novel ring-fused 2-pyridone antibiotics, Keystone Symposia: Novel Approaches Against Emerging Antimicrobial Resistance, Banff, AB, Canada. 2022.

Oral: New approaches to combat the rise of antibiotic-resistant uropathogens 4th Clinical and Scientific Advances in UTI: “Treatment and Prevention”, Virtual. 2022.

Poster: GmPcides: New class of antibiotics to kill multidrug-resistant Gram-positive pathogens International Conference on Gram-Positive Pathogens, Omaha, NE. 2022.

Oral: Mechanisms of DNA N6-methyladenosine regulation in Gram-positive bacteria Molecular Genetics of Bacteria and Phages, Madison, WI. 2021

Oral: DNA methylation from a Type I restriction modification system influences gene expression and virulence in *Streptococcus pyogenes* Molecular Genetics of Bacteria and Phages, Madison, WI. 2019.

Oral: DNA methylation promotes expression of chromosome maintenance genes. Connell Symposium, Ann Arbor, Michigan. 2019.

Poster: DNA methylation from a Type I restriction modification system influences gene expression and virulence in *Streptococcus pyogenes*, GRC Microbial Stress Response Meeting, South Hadley, MA. 2018.

Poster: N6-methyladenosine dependent regulation of developmental gene expression in *Bacillus subtilis*, Molecular Genetics of Bacteria and Phages, Madison, WI. 2017.

Poster: Identification and characterization of N6-methyladenosine DNA modifications in the *Bacillus subtilis* genome, MCDB Departmental Retreat, Ypsilanti, MI. 2016.

Oral: The *Bacillus subtilis* methylome affects gene expression. Molecular Genetics of Bacteria and Phages, Madison, WI. 2015.

Oral: Identification and characterization of N6-methyladenosine DNA modifications in the *Bacillus subtilis* genome DNA Replication and Repair Symposium, Buffalo, NY. 2015.

Poster: Detection of N6-methyladenosine and 5-methylcytidine modifications in the *Bacillus subtilis* chromosome, Molecular Genetics of Bacteria and Phages, Madison, WI. 2014.

HONORS & AWARDS

Nature’s Inspiring Women in Science: Scientific Achievement Shortlist	2022
Certificate of Excellence for contributions to MMMP immersion course	2021
Honorable mention: Nat Sternberg Thesis Prize	2021
Rackham Conference Travel Grant	2017-19
Rackham Graduate Student Research Grant	2018
Rackham Pre-Candidate Research Grant	2016
Christine Psujek Memorial Award: Best Undergraduate Thesis in Biology	2015

Poster Award, Molecular Genetics of Bacteria and Phages Meeting	2014
Otto Graf LSA Honors College Scholarship for Conference Travel	2014
MCDB Summer Research Fellowship	2014

GRANTS & FELLOWSHIPS

W.M. Keck Postdoctoral Fellowship in Molecular Medicine	2022-2023
NIH T32 Training Grant Infectious Disease/Basic Microbial Pathogenic Mechanisms	2020-2022
National Science Foundation Graduate Research Fellowship	2015-2018

MENTORSHIP

Jesús Bazan-Villicana, MSTP Student WashU	2023-Present
• Current position: Medical Student, WashU	
Morgan Timm, MSTP Student, WashU	2023-Present
• Current position: Medical Student, WashU	
Kate Wardenburg, MSTP rotation student, WashU	2021
• Current position: MSTP student, Stallings Lab	
Jennie Hazen, Ph.D., Graduate student, WashU	2020-2022
• Current position: Scientific Editor, Cleveland Clinic Lerner Research Institute	
Roberto Bloom, UMich Aspirant Summer Research Internship Program	2019

LEADERSHIP & OUTREACH

Host, UTIGA UTI Hour: “-omics in UTI”	2023
Organizer, NIH CARBIRU Trainee Coffee Hour	2023
Member, WashU DBBS Curriculum Reform Committee	2022-2023
Director, MMMP Qualification Exam (QE) Workshop	2021-2023
Organizer, Outreach Efforts with Benzie Central High School	2015-2020
• Efforts highlighted by University of Michigan Media “This is Michigan, Stories of Our State: Labs in a Box” (https://impact.govrel.umich.edu/biology-benzie)	
Organizer ‘Ask a Scientist’ event through UMich ESPA	2018
Moderator, Faculty Corner Seminar	2018
Volunteer, AWIS Young Scientists’ Expo at Forsythe Middle School	2018
Organizer, 3M Thesis Event at Benzie Central High School	2017
Organizer, University of Michigan MCDB Dept. Retreat	2017

REFERENCES

Scott Hultgren, Ph.D.
Washington University in St. Louis
Helen L. Stoeber Professor of Molecular Microbiology
Postdoctoral Advisor
hultgren@wustl.edu

Lydia Freddolino, Ph.D.
University of Michigan
Professor, Department of Biochemistry and Computational Medicine and Bioinformatics
Collaborator
lydsf@umich.edu

Lyle Simmons, Ph.D.
University of Michigan
Professor, Department of Molecular, Cellular and Developmental Biology
Graduate Advisor
lasimm@umich.edu